Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/02/2011
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Project Title:	NSCOR: Charged Particle Radiation and Resultant Oxidative Stress Elicit Deleterious Functional Changes in the Central Nervous System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Flight Program:			
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Task Description:	The proposed project will investigate responses of the mouse brain to several radiation exposure regimes corresponding to galactic cosmic rays and solar particle events. The project will characterize behavioral, electrophysiological, biochemical and structural changes both in vivo and in vitro. Radiation-elicited oxidative stress mechanisms will be investigated as important mediators of the biological responses in both wild type and mutant animals. In addition to characterization of normal tissue responses, the potential enhancement of three late risks by radiation exposure will be explored. These are seizures, ischemic injury and traumatic injury. Mathematical models of the neurotoxicity responses will developed for use in estimating risks and fitted to project-specific data and previously-published observations. The aims of the proposal will address HRP risk 1.2 and at least five of the specific gap questions identified by NASA: CNS-1, -3, -5, -6 and -11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the responses of the central nervous system to charged particle radiation has two main benefits – risk assessment and optimization of radiotherapy. The general population and aviation crews are exposed to charged particles during cross country and trans-oceanic air travel at high altitudes as well as to radon in their homes and workplaces. While these exposures are not as high as those of astronauts they are not insignificant. Cancer risks have traditionally been the focus of health risk estimates but the new awareness of sensitive targets in the brain such as neurogenic centers in the hippocampus raises question about the negative effects of such radiation on late neurodegenerative diseases and potential decline of cognitive abilities. This project directly addresses the cognitive and neurogenesis issues and indirectly addresses late neurodegenerative effects which share reactive oxygen chemistry with radiation as underlying mechanisms. Alzheimer's, Huntington's, and in particular, ALS diseases, are of interest. In the last ten years, the use of charged particles in radiation therapy has grown dramatically with proton and carbon ion treatment centers being established in the U.S., Europe, and Asia. Always limiting the effectiveness of radiotherapy is the response of normal tissue near the tumor site. The precise dose localization associated with charged particle therapy makes it ideal for tumors of the head and neck where careful avoidance of critical structures such as the optic nerve is critical. Knowledge of the dose "tolerance" of brain tissue and the reactions to relatively low doses is critical in optimizing these modalities of treatment.
Task Progress:	The CNS NSCOR focuses on the functional consequences of exposure of the central nervous system to charged particles. This work builds on previous results that demonstrated altered synaptic function, neurogenesis, microvasculature organization, and structural modification of the hippocampus. The approaches used explicitly examine the roles of reactive oxygen species and neurogenesis in mediating the effects of radiation exposure. Comparisons of responses between wild type and transgenic mice overexpressing catalase focus on the role of radiation induced hydrogen peroxide as a mediator of radiation effects. Our studies also examine the functional properties of neurons and local circuits and link these properties to the fidelity of behavioral output. Finally, aspects of radiation quality and dose rate being assessed for the endpoints above. Radiation exposure protocols include single, acute whole body exposures to protons, silicon, and iron ions. Complex exposure regimens include: 1) adaptation to priming doses of protons followed by challenge doses of iron ions and 2) dose rate will be assessed using fractionated iron ions. The animal models are 10 week old C57BL/6 males and transgenic MCAT males which over-express human catalase in mitochondria. To date we have irradiated 391 mice with protons, silicon ions and iron ions and the animals have been evaluated at 90 days post irradiation (with a small group tested at 30 days as well). In vitro models are cultured murine and human neural stem cells irradiated with several ions that are evaluated at times from days to months. Highlights from ongoing in vitro studies at U.C. Irvine (C. Limoli) are as follows: Human and rodent neural stem cells were irradiated with several ions (O, Si, Fe) at energies from 300-1000 MeV/n in the dose range 0.75 to 30 cGy. Data suggest that significant changes in oxidative stress occur that alter radiosensitivity, proliferation, and differentiation. Overall trends indicate that acute increases in ROS/RNS correlate with particle energies while longer-lived changes in oxidative stress (observed weeks to months post-irradiation) correlate with particle LET. We find that under a variety of low dose, charged particle exposures, neural stem cells exhibit a persistent increase in oxidative stress. At this point in time, precise track structure dependent trends are difficult to discern, but significant data is still forthcoming from the NSRL 11C run continuing into Nov. 2011. We will continue to define the low dose responses of neural stem cells exposed to charged particles, and attempt to elucidate if/how such changes impact redox state and their functionality in terms of survival and differentiation. The J. Fike and S. Rosi labs at UCSF analyzed tissues from animals cognitively tested at Oregon Health & Science University under the direction of Dr. Raber. The animals were injected at OHSU with BrdU four weeks before cognitive testing and tissue collection to label proliferating neural precursor cells. After animals were behaviorally tested in a contextual fear conditioning paradigm, they were sacrificed and their brain tissues were sent to UCSF where they were quantitatively assessed for neurogenesis (numbers of newly born neurons and astrocytes), numbers of activated microglia, and for neurons expressing Arc protein. The hippocampus-dependent contextual fear conditioning paradigm was used to assess cognition. No previous studies had been done assessing the relationship between the specific cognitive testing paradigm proposed for the NSCOR (contextual fear conditioning) and Arc expression. Therefore, we determined the utility of this approach in the assessment of radiation-induced injury in the brain. Initially, we analyzed a group of mice that received brain only 56Fe irradiation; doses were 0 or 1 Gy. Results from that study showed that: 1) Irradiated animals that did not receive tone-shock pairings on day 1 showed less freezing (i.e. impaired memory) and a lower fraction of Arc-expressing neurons in the free (lower) blade of the dentate gyrus (DG). This suggested reduced hippocampus-dependent spatial habituation learning, a finding that had not been previously reported. Additionally, changes in Arc expression in the free blade correlated positively with freezing; 2) Irradiated mice that did receive tone-shock pairing showed enhanced contextual freezing but a reduced percentage of Arc-positive neurons in the enclosed (upper) blade of the DG. In that case, changes in Arc expression correlated negatively with freezing. Results to date from silicon ion and proton irradiations suggest U-shaped dose response curves for contextual fear conditioning in which 0.1 to 0.25 Gy exposures produce enhanced levels of the fear response than controls or higher doses. While the relationship between Arc expression and cognition is complex, our data suggest that radiation effects on hippocampus-dependent cognition might depend upon the prominence (salience) of environmental stimuli and blade-specific Arc expression. Electrophysiology on hippocampal slices at Loma Linda University (R Vlkolinsky & I Sokolova) and UCLA (I Spigelman) is used to assess local circuit activity and changes in individual neurons. A multi-electrode system records

field potentials from groups of neurons at several locations in the hippocampus and assesses neural transmission within the tissue while patch clamp recordings of identified cells in regions CA1 and DG assess neuron function. For individual DG neurons, there was a proton dose dependent decrease in GABA mediated inhibitory currents (mIPCSs) but no other kinetic parameters were significantly affected. For excitatory activity in CA1 pyramidal neurons, proton irradiation produced decreases in amplitudes of excitatory postsynaptic currents (EPSCs) without affecting EPSC frequency and also significantly reduced transmembrane currents at certain membrane voltages. Other electrophysiological parameters were not significantly affected. Qualitatively similar alterations after silicon irradiation have been detected.

With extracellular multielectrode arrays we detected (mainly in the dose range of 0.5-1.0 Gy) reduced synaptic excitability in CA1 neurons. This effect of protons contrasts with previously described high-LET effects of 56Fe irradiation where dose-dependent increases of fEPSPs were observed at doses 1-4 Gy (head only). In the Dentate Gyrus (DG) neuronal field the effect of radiation appeared to be opposite, perhaps due to preferential effect of radiation on feed-forward inhibitory neurons that exert control over granular cell excitability. Such hypothetical effects can be well supported by our intracellular findings on reduced inhibitory synaptic currents (mIPSC) in the DG described in previous sections. Clearly, there are regional differences in radiation sensitivity to protons between CA1 and DG neurons. We also show that in MCAT mice the effects of radiation on synaptic excitability are minimized, indicating that hydrogen peroxide and perhaps other forms of reactive oxygen species are involved in mediating radiation-induced changes in the hippocampus that persist at least 3 mo post-irradiation. Effects of silicon irradiation tested in the rostral hippocampus of C57Bl/6J mice exhibit similar changes in excitability as those found in proton-irradiated mice. In the ventral hippocampus, subtle changes in dendritic excitability due to silicon irradiation were detected, but overall neuronal output of CA1 neurons was not affected. In proton irradiated mice we detected subtle changes in short-term synaptic plasticity mediated by glutamate release, but rather prominent inhibition of LTP at 0.5 and 1 Gy in CA1 neurons and at 1.0 in the DG. Again, the radiation effects were partially prevented in MCAT mice, which is suggestive of an important role of hydrogen peroxide in mediating radiation-induced changes in LTP. The changes in synaptic plasticity in normal mice may indicate a potential detrimental effect of these relatively low doses on hippocampal-dependent cognitive process due to ongoing oxidative stress. In any case, exposure to radiation appears to lead to altered information processing by this region of the brain.

Iron irradiation experiments for Aim 2 of the project will be completed in early November 2011 during the NSRL 11C run. The next set of experiments with combinations of ions will be scheduled for the summer NSRL run (NSRL 12B) for which a beam time proposal has been submitted and updates to animal protocols are in preparation.

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